JUNHUI PENG, PhD

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EDUCATION

Ph.D. University of Science and Technology of China (USTC) — Jul 2017
Computational Biology, School of Life Sciences
Advisor: Dr. Zhiyong Zhang & Dr. Yunyu Shi

B.S. University of Science and Technology of China (USTC) — Jun 2012
Biology, School of Life Sciences
Advisor: Dr. Zhiyong Zhang & Dr. Yunyu Shi

RESEARCH EXPERIENCES

Postdoctoral Researcher, Rockefeller University — Jul 2019 – Jun 2026

Advisor: Dr. Li Zhao

- Performed large scale prediction of structural interactome in *Drosophila melanogaster*, and highlighted the importance of disordered regions in proteome-wide protein-protein interactions.
- Developed a computational platform to study the origin and evolution of the interactions between sex peptide and sex peptide receptor. This demonstrates how genetic novelty gives rise to novel protein-protein interactions.
- Developed a computational pipeline to identify *Drosophila* specific *de novo* genes, revealed that *de novo* gene expression is regulated by a small number of transcription factors, and that some de novo proteins may be born well-folded.
- Demonstrated how intermolecular interactions drive protein sequence adaptative evolution in *Drosophila melanogaster* by comparative genomics analysis and populational genomics analysis.
- Collaborated with Prof. Hironori Funabiki, Prof. Kivanç Birsoy, and other colleagues in comparative genomic analysis, populational genomics analysis, and computational biology studies.

Postdoctoral Researcher, Hong Kong University of Science and Technology (HKUST) — Oct 2017 – Jun 2019

Advisor: Dr. Xuhui Huang

- Applied molecular dynamics simulations to study the dynamics of nucleosome DNA unwrapping.
- Collaborated with Prof. Benzhong Tang and other colleagues in computational biophysics studies.

Graduate studies, University of Science and Technology of China (USTC) — Sep 2012 – Jun 2017

School of Life Sciences
Advisor: Dr. Zhiyong Zhang & Dr. Yunyu Shi

- Thesis: The development and application of computational platforms that integrates biophysical data to study the structure and dynamics of biomolecules.
- Collaborated with Prof. Ming Lei, Prof. Qingguo Gong, Prof. Congzhao Zhou, and other colleagues in computational biology.

Undergraduate studies, USTC — Sep 2011 – Jun 2012

School of Life Sciences
Advisor: Zhiyong Zhang & Yunyu Shi

PUBLICATIONS *(more information at my google scholar profile)*

Postdoctoral research, Rockefeller University

1. **Peng, J.***, Zhao, L*. A predicted structural interactome reveals binding interference from intrinsically disordered regions. ***Plos Comp Biol***, 2026 **(* co-corresponding author)**
2. **Peng, J.#**, Wang, BJ.#, Svetec, N. & Zhao, L. Gene regulatory networks and essential transcription factors for de novo originated genes. ***Nat Ecol Evol***, 9, 1487–1498, 2025 **(# co-first author)**
3. Wassing, I. E., Nishiyama, A., Shikimachi, R., Jia, Q., Kikuchi, A., Hiruta, M., Sugimura, K., Hong, X., Chiba, Y., **Peng, J.**, Jenness, C., Nakanishi, M., Zhao, L., Arita, K., Funabiki, H. CDCA7 is an evolutionarily conserved hemimethylated DNA sensor in eukaryotes. ***Sci Adv***, 10(34), 2024.
4. **Peng, J.**, Svetec, N., Molina N., Zhao L. *The origin and evolution of the interactions between sex peptide and sex peptide receptor*. ***Mol Biol Evol*** msae065, 2024
5. **Peng, J.**, Zhao, L. *The origin and structural evolution of de novo genes in Drosophila*. ***Nat Commun*** 15, 2024
6. Liu Y., Liu S., Tomar A., Yen F., Unlu G., Ropek N., Weber R., Wang Y., Khan A., Gad M., **Peng J.**, et al. *Autoregulatory control of mitochondrial glutathione homeostasis*. ***Science*** 382, 820-828, 2023
7. Chung, K., Xu, L., Chai, P., **Peng, J.**, Devarkar, S. C., Pyle, A. M. *Structures of a mobile intron retroelement poised to attack its structured DNA target*. ***Science*** 378, 627-634, 2022
8. **Peng, J.**, Svetec, N., Zhao, L. *Intermolecular Interactions Drive Protein Adaptive and Coadaptive Evolution at Both Species and Population Levels*. ***Mol Biol Evol*** msab350, 2022
9. Durkin, S.M., Chakraborty, M., Abrieux, A., Lewald, K.M., Gadau, A., Svetec, N., **Peng, J.**, Kopyto, M., Langer, C.B., Chiu, J.C., et al. *Behavioral and genomic sensory adaptations underlying the pest activity of Drosophila suzukii*. ***Mol Biol Evol*** msab048, 2021

Postdoctoral research, HKUST

10. Pan, C., Liu, C., **Peng, J.**, Ren, P., and Huang, X. *Three-site and five-site fixed-charge water models compatible with AMOEBA force field*. ***J Comput Chem*** 41, 2020
11. Zhang, J., Li, A., Zou, H., **Peng, J.**, Guo, J., Wu, W., Zhang, H., Zhang, J., Gu, X., Xu, W., et al. *A "simple" donor-acceptor AIEgen with multi-stimuli responsive behavior*. ***Mater Horiz*** 7, 2020
12. Zhang, J., Liu, Q., Wu, W., **Peng, J.**, Zhang, H., Song, F., He, B., Wang, X., Sung, H.H.Y., Chen, M., et al. *Real-Time Monitoring of Hierarchical Self-Assembly and Induction of Circularly Polarized Luminescence from Achiral Luminogens*. ***ACS Nano*** 13, 2019
13. **Peng, J.**, Wang, W., Yu, Y.Q., Gu, H.L., and Huang, X. Clustering algorithms to analyze molecular dynamics simulation trajectories for complex chemical and biological systems. ***Chin J Chem Phys*** 31, 2018

Graduate research, USTC

14. **Peng, J.#**, Yuan, C.#, Hua, X., Zhang, Z*. Molecular mechanism of histone variant H2A.B on stability and assembly of nucleosome and chromatin structures*. *Epigenetics Chromatin* 13, 2020 **(# co-first author)**
15. **Peng, J.#,** Yuan, C.#, Ma, R., Zhang, Z. *Backmapping from Multiresolution Coarse-Grained Models to Atomic Structures of Large Biomolecules by Restrained Molecular Dynamics Simulations Using Bayesian Inference*. ***J Chem Theory Comput*** 15, 2019 **(# co-first author)**
16. Xu, D., Ma, R., Zhang, J., Liu, Z., Wu, B., **Peng, J.**, Zhai, Y., Gong, Q., Shi, Y., Wu, J., et al. *Dynamic Nature of CTCF Tandem 11 Zinc Fingers in Multivalent Recognition of DNA As Revealed by NMR Spectroscopy*. ***J Phys Chem Lett*** 9, 2018
17. Cheng, P.#, **Peng, J.#**, and Zhang, Z. (2017). SAXS-Oriented Ensemble Refinement of Flexible Biomolecules. Biophys J 112, **(# co-first author)**

18. Chen, C., Gu, P., Wu, J., Chen, X., Niu, S., Sun, H., Wu, L., Li, N., **Peng, J.**, Shi, S., et al. *Structural insights into POT1-TPP1 interaction and POT1 C-terminal mutations in human cancer*. ***Nat Commun*** 8, 2017
19. Xu, L., Wang, L., **Peng, J.**, Li, F., Wu, L., Zhang, B., Lv, M., Zhang, J., Gong, Q., Zhang, R., et al. *Insights into the Structure of Dimeric RNA Helicase CsdA and Indispensable Role of Its C-Terminal Regions*. ***Structure*** 25, 2017
20. **Peng, J.**, Zhang, Z. *Unraveling low-resolution structural data of large biomolecules by constructing atomic models with experiment-targeted parallel cascade selection simulations.* ***Sci Rep*** 6, 2016
21. Shao, Z., Yan, W., **Peng, J.**, Zuo, X., Zou, Y., Li, F., Gong, D., Ma, R., Wu, J., Shi, Y., et al. *Crystal structure of tRNA m1G9 methyltransferase Trm10: Insight into the catalytic mechanism and recognition of tRNA substrate*. ***Nucleic Acids Res*** 42, 2014
22. **Peng, J.**, Zhang, Z. Simulating large-scale conformational changes of proteins by accelerating collective motions obtained from principal component analysis. ***J Chem Theory Comput*** 10, 2014
23. Wen, B.#, **Peng, J.#**, Zuo, X., Gong, Q., and Zhang, Z. Characterization of protein flexibility using small-angle x-ray scattering and amplified collective motion simulations. Biophys J 107, 2014 (**# co-first author**)

FUNDING & AWARDS

Shapiro-Silverberg Fund for the Advancement of Translational Research, Rockefeller University Center for Clinical and Translational Science (RUCCTS)	2024
SMBE Travel Award, Society for Molecular Biology and Evolution	2023
C. H. Li Memorial Scholar Fund Award, Rockefeller University	2021
CAS President Award (Excellence Award), Chinese Academy of Science	2017

PRESENTATIONS

Guest Lecture, Department of Biology, Hunter College, NY	Oct 2025
Invited Seminar, Department of Biology, Rutgers University–Camden, NJ	Feb 2025
Contributed, SMBE Satellite Meeting on De Novo Gene Birth, Texas A&M University, TX	Nov 2023
Contributed, the 64th Annual Drosophila Research Conference, Chicago, IL	Mar 2023
Contributed, 2023 New York Area Population Genomics Meeting, NY	Jan 2023
Contributed, BSVMRKYZ Super-Group Meeting, Rockefeller University, NY	Nov 2022
Contributed, Pels Family Center Chemical and Structural Biology Retreat, NY	Oct 2022
Poster, virtual, the 63rd Annual Drosophila Research Conference, CA	Apr 2022
Contributed, virtual, the 2nd AsiaEvo Conference, Japan	Aug 2021
Contributed, Chinese Society of Biochemistry and Molecular Biology Conference, China	Aug 2014

TEACHING & TRAINING

Guest Lecturing at Hunter College Department of Biology’s Fall 2025 Seminar Series — 2025
With support from Hunter and RockEDU Science Outreach

Tri-I institute Teaching Course — 2025
Training program for postdocs on teaching and mentoring career paths

SERVICE & OUTREACH

Faculty Search Ambassador — 2023

Navigate candidate through campus and introduce them to faculty members and resource centers, Rockefeller University Faculty Search Seminar 2023

Science Networking Host 2022

Host of virtual networking at 2022 Annual Drosophila Meeting, session of Evolution, Immunity, and the Microbiome

Ad Hoc Reviewer 2017 – Present

Molecular Biology and Evolution, PLOS Genetics, Cell Reports, Genome Biology and Evolution, Journal of Molecular Evolution, Journal of Evolutionary Biology, Biology of Reproduction, PLOS One, Archives of Biochemistry and Biophysics, BioSystems, Computational Biology and Chemistry, Journal of Molecular Structure